If $\mathcal{F}$ and $\mathcal{G}$ are both fans in the same space $\mathbb{R}^d$, then we define their *common refinement* as

$$\mathcal{F} \wedge \mathcal{G} \quad := \quad \{C \cap C' : \ C \in \mathcal{F}, \ C' \in \mathcal{G}\}.$$

If $\mathcal{F}$ is a fan in $\mathbb{R}^d$ and $V \subseteq \mathbb{R}^d$ is a vector subspace, then the *restriction* of $\mathcal{F}$ to V is the fan

$$\mathcal{F}\Big|V \quad := \quad \{C \cap V : \ C \in \mathcal{F}\}.$$

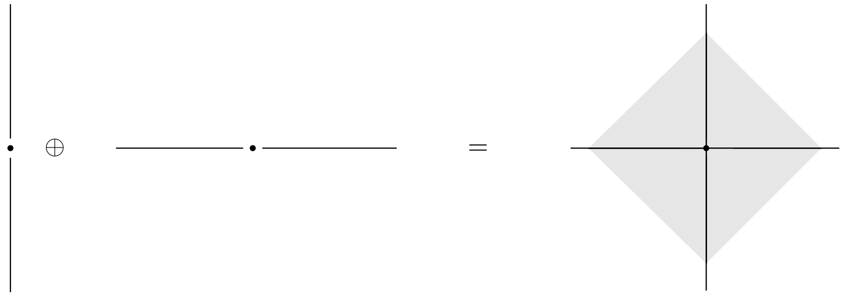
It is easy to check that all three constructions are again fans in the sense of Definition 7.1. If you want, you can consider $\mathcal{G}_V := \{V\}$ as a (noncomplete) fan in $\mathbb{R}^d$: then the restriction to V is the intersection with $\mathcal{G}_V$, that is, $\mathcal{F}\Big|V = \mathcal{F} \wedge \mathcal{G}_V$.

Lemma 7.7. *Let $P \subseteq \mathbb{R}^p$ and $Q \subseteq \mathbb{R}^q$ be two polytopes. Then the normal fan of the product $P \times Q \subseteq \mathbb{R}^{p+q}$ is the direct sum*

$$\mathcal{N}(P \times Q) \quad = \quad \mathcal{N}(P) \oplus \mathcal{N}(Q). \qquad \square$$

Example 7.8. The normal fan of the cube $C_d = \{\boldsymbol{x} \in \mathbb{R}^d : -1 \leq x_i \leq 1\}$ coincides with the face fan of its polar, the d-crosspolytope (cf. Exercise 7.1 for the general case of this).

We find that the normal fan is given by the arrangement of coordinate hyperplanes in $(\mathbb{R}^d)^*$. This is quite trivial (since it is trivial to optimize over a cube), and it also confirms Lemma 7.7: the fan of the arrangement of coordinate hyperplanes is the direct sum of 1-dimensional fans.



Also, the cones in the fan are characterized by the sign function: each cone in the fan can be identified with a vector in $\{+, -, 0\}^d$, and the orthants correspond to the sign vectors in $\{+, -\}^d$.

7.2 Projections and Minkowski Sums

Let $P \subseteq \mathbb{R}^p$ be a p-polytope, and let $\pi : \mathbb{R}^p \longrightarrow \mathbb{R}^d$ be an *affine map*,

$$\pi(\boldsymbol{x}) = A\boldsymbol{x} - \boldsymbol{z},$$

with $A \in (\mathbb{R}^d)^p = \mathbb{R}^{d\times p}$ and $\boldsymbol{z} \in \mathbb{R}^d$.

If π is injective (that is, A has rank p), then we refer to it as an *affine transformation*, and $\pi(P)$ is affinely isomorphic to P.

If π is not required to be injective, then we refer to it as an *affine projection* or an *affine map*. In this case $Q := \pi(P)$ is again a polytope, whose dimension is $\dim(Q) = \dim(\pi(\mathbb{R}^p)) = \operatorname{rank}(A)$, where we had assumed $\dim(P) = p$.

We usually assume that π is surjective — we may do this, after restricting the image of π to $\pi(\mathbb{R}^p) \subseteq \mathbb{R}^d$ — and so π maps the p-polytope $P \subseteq \mathbb{R}^p$ to a d-polytope $Q \subseteq \mathbb{R}^d$. Also, if we are only interested in properties that are invariant under translation, then we may assume that $\boldsymbol{z} = \boldsymbol{0}$ and that the map π is actually linear.

Definition 7.9. A *projection of polytopes*

$$\pi:\ P \longrightarrow Q$$

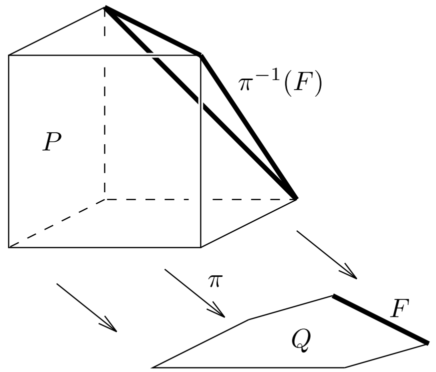
is an affine map $\pi : \mathbb{R}^p \longrightarrow \mathbb{R}^d$, $\boldsymbol{x} \longmapsto A\boldsymbol{x} - \boldsymbol{z}$, such that $P \subseteq \mathbb{R}^p$ is a p-polytope, $Q \subseteq \mathbb{R}^d$ is a d-polytope, and $\pi(P) = Q$.

Here is a very simple, but basic, fact about projections.

Lemma 7.10. *Let $\pi : P \longrightarrow Q$ be a projection of polytopes. Then for every face of Q, $F \in L(Q)$, the preimage $\pi^{-1}(F) = \{\boldsymbol{y} \in P : \pi(\boldsymbol{y}) \in Q\}$ is a face of P.*

Furthermore, if F, G are faces of Q, then $F \subseteq G$ holds if and only if $\pi^{-1}(F) \subseteq \pi^{-1}(G)$.

Proof. If $\boldsymbol{c} \in (\mathbb{R}^d)^*$ defines F, then $\boldsymbol{c} \circ \pi$ defines $\pi^{-1}(F)$. Instead of the corresponding trivial calculation, we give a picture.



□

The linear algebra behind this construction is quite simple. From the surjective map $\pi : \mathbb{R}^p \longrightarrow \mathbb{R}^d$ we get a dual map $\pi^* : (\mathbb{R}^d)^* \longrightarrow (\mathbb{R}^p)^*$, mapping $\boldsymbol{c} \longmapsto \boldsymbol{c} \circ \pi$, which is injective. This distinguishes a certain set of functions on $\mathbb{R}^p$ (and thus on $\mathbb{R}^d$): those that are constant on the fibers of π. The embedding π^* is used in the following lemma.

Lemma 7.11. *The normal fan $\mathcal{N}(Q)$ of a projected polytope Q is isomorphic, via π^*, to the restriction of $\mathcal{N}(P)$ to the image of π^*, the linear subspace $\pi^*(\mathbb{R}^d)^*$:*

$$\mathcal{N}(Q) \quad \overset{\pi^*}{\cong} \quad \mathcal{N}(P)\Big|\pi^*(\mathbb{R}^d)^*.$$

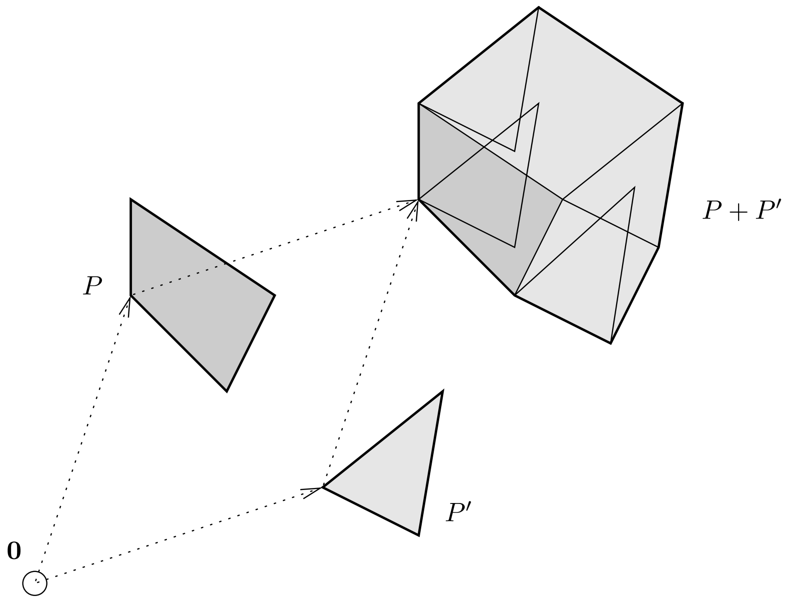
We had considered very special projection maps in Lecture 1: projections along a coordinate axis (with $d = p - 1$), which correspond to Fourier-Motzkin elimination. The general case is certainly interesting. In fact, if we try to view polytopes as a category in natural terms, then probably isomorphism should be affine isomorphism, and surjective maps should be affine projection maps. The geometry of projections is not really understood: we will talk about this later.

In terms of polarity, which links the two versions of Fourier-Motzkin elimination given in Lecture 1, note that the polar operation to projection is intersection, taking $Q = P \cap V$, where $\mathbf{0} \in \operatorname{relint}(P)$ and $V \subseteq \mathbb{R}^d$ is a vector subspace. In this case we see that $\mathcal{F}(Q) = \mathcal{F}(P)\big|V$.

A simple application of projection is the construction of Minkowski sums, which we already met in Section 1.1. Here we work out only the case of two summands, as the extension to more summands is then obvious.

The *Minkowski sum* (or *vector sum*) of two polytopes P and P' in $\mathbb{R}^d$ is

$$P \ + \ P' \quad := \quad \{\boldsymbol{x} + \boldsymbol{x}' : \boldsymbol{x} \in P,\ \boldsymbol{x}' \in P'\}.$$



We use the projection map $\pi : \mathbb{R}^{2d} \longrightarrow \mathbb{R}^d$ given by $\pi\binom{\boldsymbol{x}}{\boldsymbol{x}'} := \boldsymbol{x}+\boldsymbol{x}'$, whose dual map $\pi^* : (\mathbb{R}^d)^* \longrightarrow (\mathbb{R}^{2d})^*$ is the diagonal map $\pi^*(\boldsymbol{c}) = (\boldsymbol{c},\boldsymbol{c})$, by

$$\pi^*(\boldsymbol{c})\begin{pmatrix}\boldsymbol{x}\\ \boldsymbol{x}'\end{pmatrix} \;=\; \boldsymbol{c}\left(\pi\begin{pmatrix}\boldsymbol{x}\\ \boldsymbol{x}'\end{pmatrix}\right) \;=\; \boldsymbol{c}(\boldsymbol{x}+\boldsymbol{x}') \;=\; (\boldsymbol{c},\boldsymbol{c})\begin{pmatrix}\boldsymbol{x}\\ \boldsymbol{x}'\end{pmatrix}.$$

Now we can write the Minkowski sum as a projection of the product:

$$P \;+\; P' \;\;:=\;\; \pi(P \times P').$$

Thus, putting together very simple observations, we get the normal fan of a Minkowski sum. (Careful: it is *not* the direct sum of the fans of the factors!)

Proposition 7.12. *The normal fan of a Minkowski sum is the common refinement of the individual fans:*

$$\mathcal{N}(P+P') \;\;=\;\; \mathcal{N}(P) \;\wedge\; \mathcal{N}(P').$$

Proof.

$$\begin{aligned}\mathcal{N}(P+P') \;\;&=\;\; \mathcal{N}(\pi(P\times P')) \;\;\stackrel{\pi*}{\cong}\;\; \mathcal{N}(P\times P')\Big|\pi^*(\mathbb{R}^d)^* = \\ &=\;\; (\mathcal{N}(P)\oplus\mathcal{N}(P'))\Big|\pi^*(\mathbb{R}^d)^* \;\;\cong\;\; \mathcal{N}(P) \;\wedge\; \mathcal{N}(P'),\end{aligned}$$

using Lemma 7.7 for the direct sums, Lemma 7.11 for the projection, and the dual map $\pi^*(\boldsymbol{c}) = (\boldsymbol{c},\boldsymbol{c})$. □

7.3 Zonotopes

Zonotopes are special polytopes that can be viewed in various ways: for example, as projections of cubes, as Minkowski sums of line segments, and as sets of bounded linear combinations of vector configurations. Each of these descriptions gives different insight into the combinatorics of zonotopes. The following includes several such descriptions, all of which lead us to the same "associated" system of sign vectors that describes the combinatorics of a zonotope. The main goal will be to see in what sense zonotopes and arrangements can be considered equivalent, and how the combinatorial structure of a zonotope is given by an oriented matroid.

After the general discussion of projections in the last section we now consider a very special (but interesting) case: projections of cubes, that is, $\pi : P \longrightarrow Q$, $\boldsymbol{x} \longmapsto V\boldsymbol{x}+\boldsymbol{z}$ is an arbitrary (surjective) affine map, but P is the d-cube,

$$P \;=\; C_p \;=\; \{\boldsymbol{x}\in\mathbb{R}^p : -1 \le x_i \le 1 \text{ for all } i\}.$$